

Akanksha Sachan

CONTACT

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Aging Institute & Dept. of Immunology, School of Medicine, UPitt [LinkedIn](#)
Machine learning for Genomics & Epigenetics [Twitter](#) | [Github](#) | [Google Scholar](#)

EDUCATION

University of Pittsburgh '24 – Present
PhD Candidate in Computational Biology
Advised by Dr. Aditi Gurkar & Dr. Jishnu Das

Carnegie Mellon University '22 – '24
PhD Student in Computational Biology
Comp-Bio Department, School of Computer Science | GPA: 3.58/4

Indian Institute of Technology (IIT), Bombay '18 – '22
BTech. in Chemical Engineering | GPA: 8.06/10

AWARDS AND HONORS

American Federation for Aging Research (AFAR) Graduate Scholarship '25
Best Poster Award at the Aging Institute Research Day, UPitt '24
1st Place, Student A100 track NeurIPS LLM Efficiency Challenge | team of 5 '23
\$500 Track Prize (Healthcare Accessibility for Underrepresented Communities), Pitt Challenge Hackathon (6th annual) | team of 3 '22
Recipient of **Undergraduate Research Award** (URA-01) by IIT Bombay '22
Recipient of Merit-cum-Means (MCM) Scholarship by IIT Bombay for two semesters for financial hardship '21
Top 0.2% (Rank 3188 out of **1.1M+** candidates) in JEE (Main) '18
Top 1.5% (Rank 2517 out of **160k+** candidates) in JEE (Advanced) '18

INVITED TALKS

“FOCAL-FIRE: Factorized Interpretable Regulatory Encodings” | *paper-in-prep*
International Society for Computational Biology (ISCB)/(ISMB) Jul '26, '25
Keystone Symposia: Artificial Intelligence in Molecular Biology Sep '25
American Association of Immunologists (AAI) Annual Meeting 2026 Apr '26
Center of Lung Aging and Regeneration, School of Medicine, UPitt Nov '25
30th Annual School of Medicine Graduate Student Research Symposium, UPitt Oct '25
Joint CMU-UPitt Ph.D. Program in Computational Biology, Orientation Retreat Aug '25
“Single-nuclei RNA-seq reveals sex-dimorphic metabolic rewiring caused by accelerated aging” | *submitted to Nature Aging*
Center for Systems Immunology Annual Retreat, UPitt Nov '25
23rd Annual Department of Immunology Scientific Retreat, UPitt Sep '25
Aging Institute Research Day, UPitt Nov '24
“Micro-MPA digital twin, with sclerostin-dependent RANKL production reproduces pivotal dual-action effect of romosozumab found in clinical trials”
50th European Calcified Tissue Society (ECTS) Congress Apr '23

**RESEARCH
EXPERIENCE
PRIOR TO THESIS**

Mapping the epigenetic landscape of single-cells using 3-D chromatin architecture	'22 – '24
<i>Supervisor – Prof. Jian Ma, Computational Biology Department, SCS, CMU</i> Developed a framework to identify cis-regulatory 3-D genomic hubs controlling target-gene expression and key transcription factors in healthy and cancerous mouse-brain single-cell, multi-omic profiling.	
Translational Mechanobiology Lab, MBI, National University of Singapore	'22
<i>Summer Internship Supervisor – Prof. Hanry Yu, Dept. of Physiology, YLL School of Medicine</i> Built a Generative-Adversarial-Network (GAN) pipeline for colorectal-cancer imaging to support clinical deep-learning deployment	
Lab for Bone Biomechanics, Institute of Biomechanics, ETH Zurich	'21
<i>Summer Internship Supervisor – Prof. Ralph Muller, Dept. of Health Sci&Tech</i> Added a Type-2 Diabetes pathway to an agent-based osteoporosis model and ran large-scale simulations on the CSCS Swiss National Supercomputing Center	
Multicellular Systems Engineering Lab, University of Notre Dame	'20
<i>Summer Internship Supervisor – Prof. Jeremiah Zartman, Chemical & Biomolecular Engineering</i> Built a morphometric feature-extraction pipeline for <i>Drosophila</i> wing imaginal discs (OpenCV, Fiji) to study organism-wide cell-signaling	

**KEY
COURSEWORK**

Machine Learning & Statistics: Intermediary Statistics (CMU: 36-705), Probabilistic Graphical Models (CMU: 10-708), Introduction to Machine Learning (CMU: 10-701)	
Predicting genomic-hub mediated TF-perturbation response via geometric deep learning	'23
<i>10-708: Probabilistic Graphical Models</i> Augmented a GNN for Perturb-seq response prediction with a Hi-C contact graph (Cancer cell-line), adding 3D-genome topology to the co-expression and Gene Ontology graph features.	
Single-cell DNA methylation prediction using single-cell HiC data and Graph NNs	'23
<i>02-710: Computational Genomics</i> Developed a GNN to predict methylation from HiC data, connecting structure to phenotype; clustered single-nucleus DNA methylation data with K-means and t-SNE.	

**ATHLETIC
ACCOLADES**

Bronze Medal 35th Intercollegiate (inter-IIT) Aquatics Meet 3rd in Medley-Relay	'19
3rd Position Comp-Bio Department, Pretty Good Race (5km) for SCS, CMU	'23
Swam 12-hours Swimathon organized by the IIT Bombay aquatics committee	'19
Ran 26.2 miles Dick's Sporting Goods Pittsburgh Marathon 5 hours 15 mins	'24
Ran 13.1 miles Dick's Sporting Goods Pittsburgh Half Marathon 2 hours 30 mins	'23

**LEADERSHIP &
SOCIETAL
CONTRIBUTIONS**

President • Vice President

Aug '25 – Present

Allegheny Science Policy and Governance (ASPG)

Leading a recognized chapter of the **National Science Policy Network**
Bridged scientists and PA legislators on policies about **AI/data-centers**
and **Aging** at the State Capitol, Harrisburg, PA

Technights Coordinator

Aug '23 – Aug '25

Carnegie Mellon University, Robotics Institute

Organized and led a **computer-science outreach** program teaching
middle-school girls sessions on foundation models, and deep learning.

NIH Consortium Hackathon, Co-organizer & Team Lead

Mar '24

4D Nucleome, Seattle, Washington

Led a team of 5 to build integrative pipelines for single-cell HiC analysis
from cell-atlas-scale datasets

Reviewer

ICML Foundation Model for Life Sciences Workshop 2026, PNAS Journal
(assisted PI), *Aging Cell Journal* (assisted PI)

Teaching Assistant

Fall '23 – Spring '24

02-760: Lab Methods for Computational Biologists, SCS, CMU

Taught a lecture on **genomics technologies** to a class of ~20 students,
revamping the RNA sequencing module of the course.

Treasurer • Senator

Aug '23 – Aug '25

Graduate Student Association (Joint CMU-UPitt Comp-Bio PhD Program)

Editorial Board Member • Design Convenor

'19 – '21

Insight, the student newsletter of IIT Bombay

Convenor, Girls Swim Team

'19 – '20

Aquatics, Indian Institute of Technology Bombay

**CONSORTIUMS &
WORKSHOPS**

NIH Consortium Annual Meetings

'23, '25

IGVF, SenNet:Cellular Senescence Network, 4D Nucleome

Grace Hopper Celebration

Oct '24

AnitaB.org, Philadelphia

Computational Genomics Summer Institute

Jul '23

University of California, Los Angeles

Grad Cohort for Women

April '23

Computing Research Association (CRA-WP), San Francisco

**COMPUTE
ALLOCATION
GRANTS**

ACCESS (NSF Advanced Cyberinfrastructure Coordination Ecosystem:
Services & Support) **1.5M** credits on the **NCSA Delta** GPU supercomputer

'26